



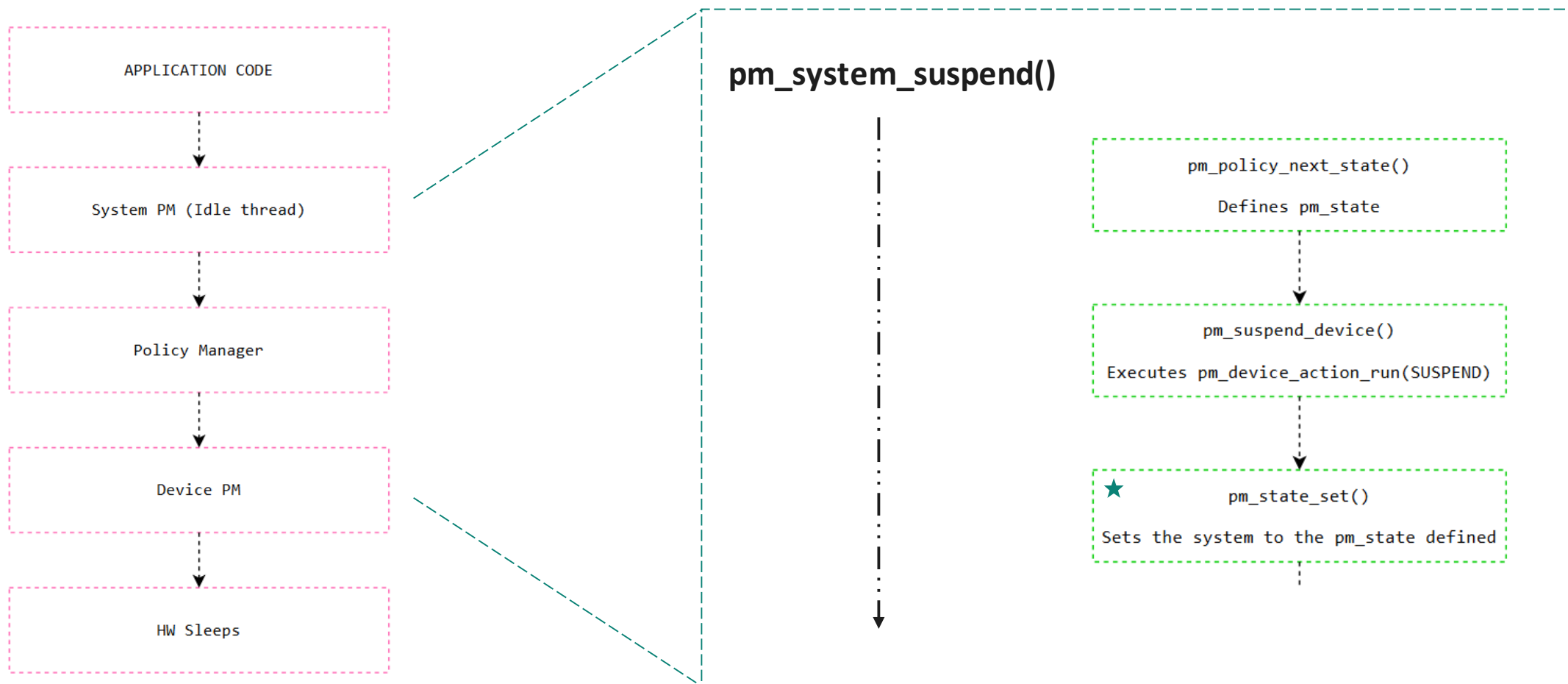
SMIF Power Management in Zephyr

Dylan Arturo Velasco Pinilla

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Zephyr Power Mangament Flow



★ **Vendor-specific implementation**

Based on **Infineon's** System Power Management

Device Driver

The Zephyr driver requires three specific components to enable power management

- **Action Callback** (`pm_device_action_cb_t()`)
- **Device PM context**: Pointer to callback
- **Zephyr registration with a pointer to the PM context**: Device specific instantiation.

```
ifx_serial_memory_flash_pm_callback(cy_stc_syspm_callback_params_t *callbackParams,
cy_en_syspm_callback_mode_t mode)
{
    if (mode == CY_SYSPM_CHECK_READY) {
        smif0_crypto_input1 = SMIF_CRYPTO_INPUT1(SMIF0_CORE); }
    if (mode == CY_SYSPM_AFTER_TRANSITION) {
        SMIF_CRYPTO_INPUT1(SMIF0_CORE) = smif0_crypto_input1; }
}
```

Device Driver

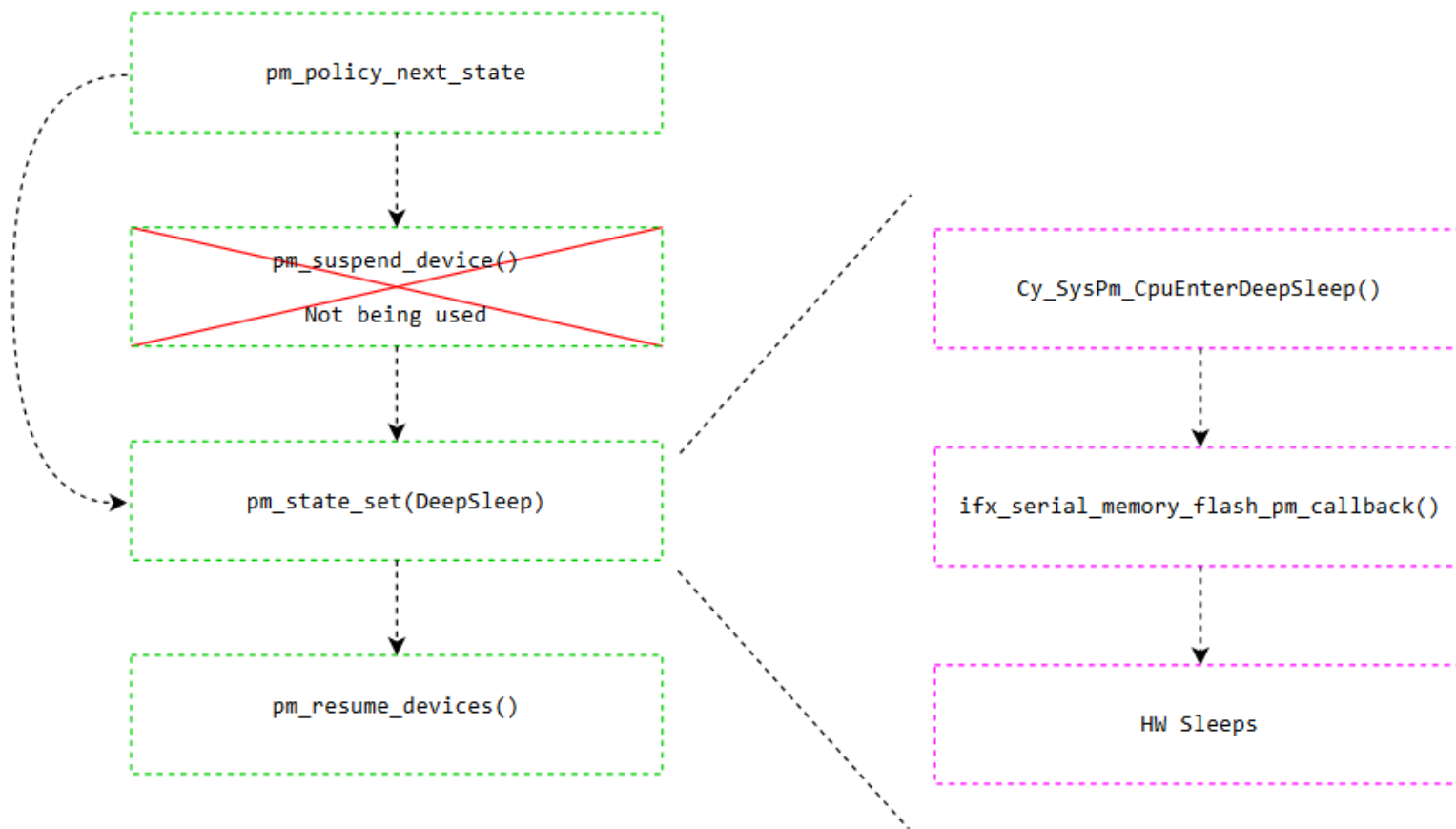
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```
cy_stc_syspm_callback_t flash_deep_sleep = {&ifx_serial_memory_flash_pm_callback,  
                                           CY_SYSPM_DEEPSLEEP,  
                                           CY_SYSPM_SKIP_BEFORE_TRANSITION};
```

```
DEVICE_DT_INST_DEFINE(0, ifx_serial_memory_flash_init, NULL, &flash_data, &flash_config,  
                      POST_KERNEL, CONFIG_FLASH_INIT_PRIORITY, &ifx_serial_memory_flash_driver_api);
```

Current Implementation



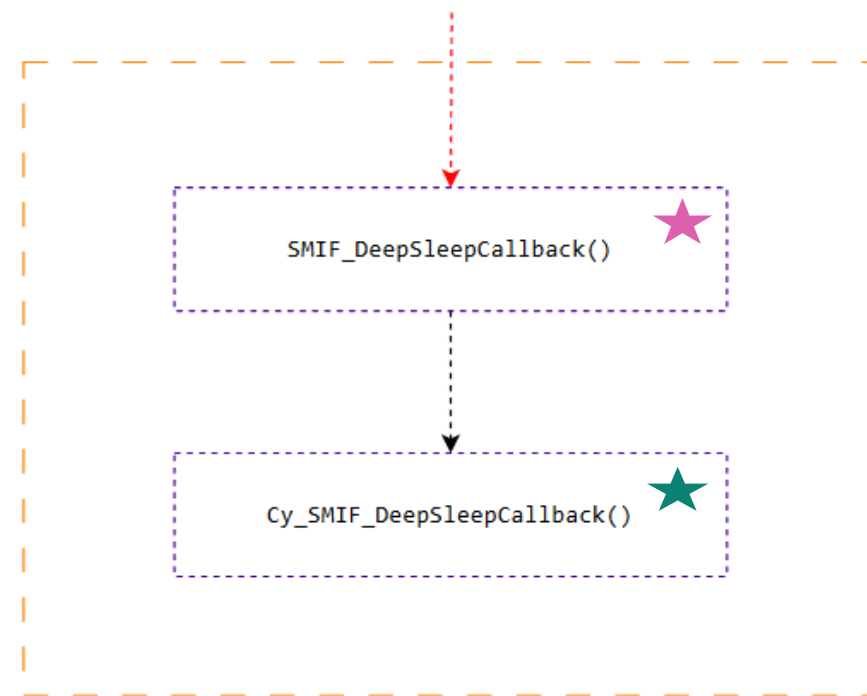
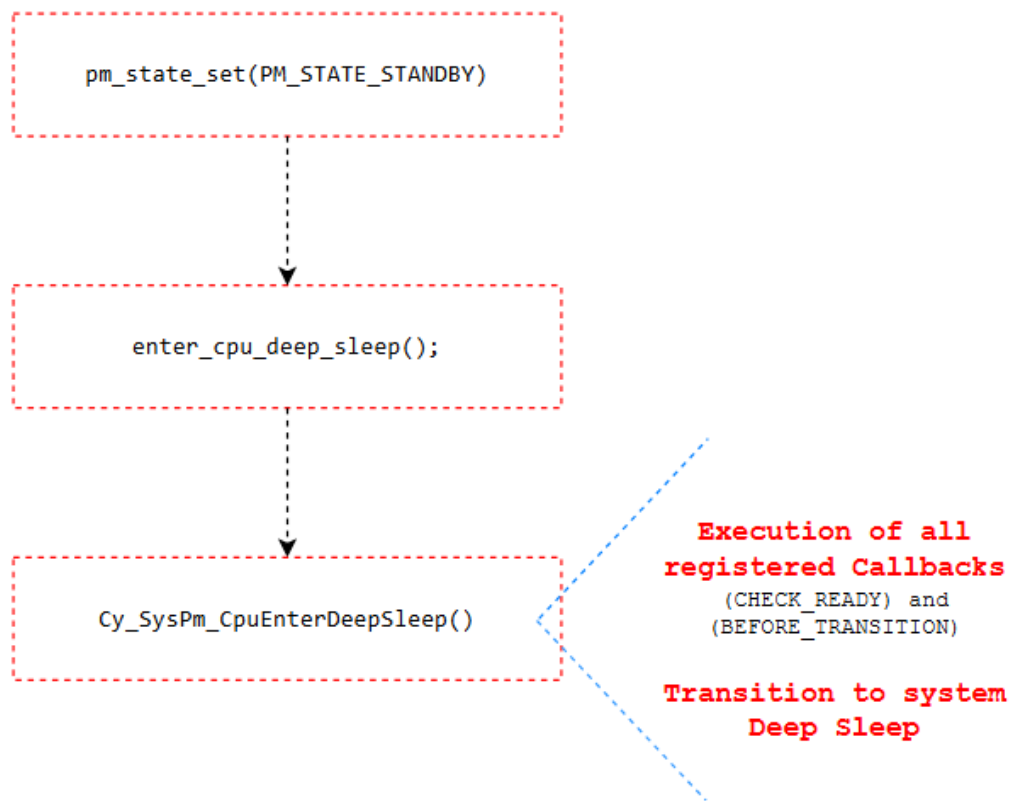
- ❖ **Managed by Infineon's PM subsystem, not Zephyr's:**
- ❖ **Callback is registered in Infineon's PM subsystem**
- ❖ **Zephyr device instantiation occurs without PM device management**

Solution Paths

Two Approaches

- **Create** a Zephyr-specific SMIF driver callback
- **Modify** the SMIF Infineon-specific template callback

Second Approach: PM_state_set() Infineon Implementation



- ★ **Template API:** Handles the transition of the SMIF into and out of Deep Sleep mode.
- ★ **Custom** external memory low power state handler

Cy_SMIF_DeepSleepCallback()

```
case CY_SYSPM_CHECK_READY:
```

```
{
```

```
    /* Check if API is not busy executing transfer operation */
```

```
    /* Add check memory that memory not in progress */
```

```
case CY_SYSPM_BEFORE_TRANSITION:
```

```
{
```

```
    /* This code executes inside critical section and enabling active
```

```
    /* interrupt source makes interrupt pending in the NVIC.
```

```
    /* Put external memory in low power mode */
```

```
case CY_SYSPM_AFTER_TRANSITION:
```

```
{
```

```
    /* Restore content of crypto registers */
```

```
    /* Put external memory in active mode */
```

SMIF_DeepSleepCallback()

– Infineon Quad-SPI NOR Flash – Power Modes:

- Active
- Stand by
- Deep power down

– Callback implementation:

CY_SYSPM_CHECK_READY: Verify memory is in **Standby** (no internal operations in progress)

CY_SYSPM_BEFORE_TRANSITION: Send **ENDPD_0_0** command via SMIF to enter **Deep Power Down**

CY_SYSPM_AFTER_TRANSITION: Wake memory from DPD via **CS# pulse**

